A First Course in Abstract Algebra

Section 25: Ordered Rings and Fields

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

25 Ordered Rings and Fields 3

Exercise 25.1: The high-degree ordering of monomials 3

Exercise 25.2: Laurent monomials in the low-degree ordering 3

Exercise 25.3: Positive elements in the conjugate ordering 3

Exercise 25.4: Ordering five polynomials, set 4 3

Exercise 25.5: Ordering five polynomials, set 5 4

Exercise 25.6: Ordering five polynomials, set 6 4

Exercise 25.7: Ordering five polynomials, set 7 4

Exercise 25.8: Ordering five polynomials, set 8 5

Exercise 25.9: Ordering five polynomials, set 9 5

Exercise 25.10: Ordering Laurent monomials with coefficients 5

Exercise 25.11: Ordering five rational Laurent series 5

Exercise 25.12: Comparing equal constant terms in Laurent series 6

Exercise 25.13: A second comparison of equal constant terms 6

Exercise 25.14: An ordered field containing nonreal numbers 6

Exercise 25.15: True or false: orderings 6

Exercise 25.16: Making pi larger than every rational 7
Exercise 25.17: Conjugation of a quadratic ring 7
Exercise 25.18: Cone membership means greater than zero 7
Exercise 25.19: Equal positive multiples 7
Exercise 25.20: Reversing signs reverses an inequality 8
Exercise 25.21: A negative times a positive 8
Exercise 25.22: A positive quotient 8
Exercise 25.23: Inverting a positive element below one 8
Exercise 25.24: Inverting a negative element above minus one 8
Exercise 25.25: Transporting an ordering across an isomorphism 8
Exercise 25.26: The induced ordering on a subring 8
Exercise 25.27: Recovering the cone from the order relation 9
Exercise 25.28: Injectivity of an odd power in an ordered domain 9
Exercise 25.29: Eight different orderings in two variables 9

25 Ordered Rings and Fields

Exercise 25.1: The high-degree ordering of monomials

For P_{high}, the sign of a nonzero polynomial is the sign of its highest-degree coefficient. Thus $x - a$ is positive for every $a > 0$ in R, and $x^{n+1} - x^n$ is positive for every $n \geq 1$. The requested ordering is

$$\boxed{0 < a < x < x^2 < x^3 < \cdots < x^n < x^{n+1} < \cdots}.$$

The comparison $a < x$ concerns every positive constant, regardless of its size in the original ring.

Exercise 25.2: Laurent monomials in the low-degree ordering

The sign of a nonzero Laurent series is determined by the coefficient at its least exponent. For integers $i < j$, the difference $x^i - x^j$ has least term x^i with positive coefficient. Consequently $x^j < x^i$. All monomials are positive, and the full order is

$$\boxed{0 < \cdots < x^3 < x^2 < x < 1 < x^{-1} < x^{-2} < x^{-3} < \cdots}.$$

In particular x is positive and smaller than every positive constant of the coefficient field.

Exercise 25.3: Positive elements in the conjugate ordering

The ordering is transported by $\phi(m + n\sqrt{2}) = m - n\sqrt{2}$. Therefore an element is positive in this new ordering exactly when $m - n\sqrt{2} > 0$ in the usual real order. In integer-coefficient cases this means:

- if $n = 0$, then $m > 0$;
- if $n > 0$, then $m > 0$ and $m^2 > 2n^2$;
- if $n < 0$, then either $m \geq 0$, or $m < 0$ and $2n^2 > m^2$.

Squaring in these comparisons is valid only after the relevant sides have been made nonnegative. For example $-\sqrt{2}$ has $m = 0, n = -1$, so it is positive in this ordering.

Exercise 25.4: Ordering five polynomials, set 4

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{a < c < d < e < b}$. The deciding terms of its four successive differences are

$$5, 2x, 2, 3.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{d < b < a < e < c}$. The deciding terms of its four successive differences are

$$7x^2, 6x, 4x^2, 3x^2.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.5: Ordering five polynomials, set 5

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{a < c < e < d < b}$. The deciding terms of its four successive differences are

$$1, 5x, 8x^2, 3x.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{e < c < b < a < d}$. The deciding terms of its four successive differences are

$$4x^5, 11x^4, 8x^3, x^5.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.6: Ordering five polynomials, set 6

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{c < a < b < e < d}$. The deciding terms of its four successive differences are

$$2, 3, 2x, 6x^3.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{e < c < a < b < d}$. The deciding terms of its four successive differences are

$$5x^5, 5x^2, x^3, 8x^4.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.7: Ordering five polynomials, set 7

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{d < a < b < c < e}$. The deciding terms of its four successive differences are

$$3x, 2x^2, 2x, x^2.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{d < c < e < a < b}$. The deciding terms of its four successive differences are

$$x^2, x^2, 5x^3, 4x^4.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.8: Ordering five polynomials, set 8

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{e < a < c < b < d}$. The deciding terms of its four successive differences are

$$x, 3x^2, 2x^2, x.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{c < d < a < e < b}$. The deciding terms of its four successive differences are

$$x, 6x^3, x^2, 4x^2.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.9: Ordering five polynomials, set 9

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{c < a < e < d < b}$. The deciding terms of its four successive differences are

$$x^3, 6x^2, 4x^3, 2.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{e < c < b < a < d}$. The deciding terms of its four successive differences are

$$8x^3, x^3, x, 4x^4.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.10: Ordering Laurent monomials with coefficients

The least exponent controls the sign. The two negative terms come first: $-5x^{-2} < -3x^{-2}$. Among the positive terms, $4x < x^{-1} < 2x^{-1}$, because a positive x^{-1} term exceeds any positive multiple of x, and the coefficient 1 is less than 2 at exponent -1. Thus the increasing label order is $\boxed{b < d < e < a < c}$.

Exercise 25.11: Ordering five rational Laurent series

Use formal geometric-series division to expose each least term:

$$a = 1 + x + \cdots, \quad b = x^2 + \cdots, \quad c = x^{-1} + 1 + \cdots,$$
$$d = -x + \cdots, \qquad e = \tfrac{3}{4}x^{-1} - \tfrac{1}{2} + \cdots.$$

The negative series d comes first. The positive infinitesimal b is below a, whose constant term is 1. Both e and c have positive exponent--1 terms, so exceed a; their coefficients compare as $3/4 < 1$. Therefore $\boxed{d < b < a < e < c}$. The dots hide higher exponents, which cannot alter these deciding terms.

Exercise 25.12: Comparing equal constant terms in Laurent series

Expansion gives

$$a = \tfrac{5}{3}x^{-1} - \tfrac{26}{9} + \cdots, \qquad b = -\tfrac{1}{2} + \tfrac{5}{8}x + \cdots, \qquad c = \tfrac{7}{4} + \cdots,$$

$$d = \tfrac{9}{2} + \cdots, \qquad e = -\tfrac{1}{2} + \tfrac{2}{3}x + \cdots.$$

For b and e, the constant terms cancel; the coefficient of x in $e - b$ is $2/3 - 5/8 = 1/24 > 0$, proving $b < e$. Their negative constant terms put both below c and d. Since $7/4 < 9/2$, $c < d$, and the positive x^{-1} term puts a above them all. Thus $\boxed{b < e < c < d < a}$.

Exercise 25.13: A second comparison of equal constant terms

Formal expansions yield

$$a = 1 - 2x + \cdots, \quad b = 1 - \tfrac{10}{3}x + \cdots, \quad c = \tfrac{1}{4}x^{-1} - \tfrac{1}{16} + \cdots,$$

$$d = -\tfrac{1}{3}x^{-1} - \tfrac{1}{9} + \cdots, \qquad e = 4x + \cdots.$$

The negative x^{-1} term places d first, and the positive such term places c last. The positive infinitesimal e lies below the two series with constant term 1. Finally $a - b$ has least term $(4/3)x > 0$, so $b < a$. The increasing order is $\boxed{d < e < b < a < c}$.

Exercise 25.14: An ordered field containing nonreal numbers

Let $K = \mathbb{Q}(\sqrt[3]{2}) \subset \mathbb{R}$, with its induced real ordering, and let $L \subset \mathbb{C}$ be the isomorphic field in the question, containing $\sqrt[3]{2}(-1 + i\sqrt{3})/2$. If $\phi : K \to L$ is the stated isomorphism, declare $\phi(a) >_L 0$ exactly when $a > 0$ in K. The cone axioms transfer through the isomorphism, so this orders L, although L contains nonreal complex numbers. There is no contradiction: the order uses the algebraic field structure of L and is not a restriction of an order on all of $\mathbb{C}$. The obstruction $i^2 = -1$ rules out ordering $\mathbb{C}$ itself, not every nonreal subfield.

Exercise 25.15: True or false: orderings

(a) True: $1 = 1^2$ is positive; positive integer sums are then positive, and trichotomy forces the negatives to be negative.

(b) True: every usual positive real is a nonzero square, so must be positive in any field ordering. Trichotomy fixes all signs.

(c) False: $\mathbb{Q}(\sqrt{2})$ has its usual ordering and the ordering transported by conjugation $\sqrt{2} \mapsto -\sqrt{2}$, which disagree on $\sqrt{2}$.

(d) True: positive integers are positive, their inverses are positive, and hence all usual positive rationals are positive. Trichotomy leaves no alternative.

(e) True: either the low- or high-degree coefficient ordering extends the given constant ordering.

(f) False: without $a > 0$, take $a = 0, b = 1$; no positive n has $1 < n0$.

(g) True: this is equivalent to the Archimedean definition. For $b > 0$ it is that definition; for $b \leq 0$, choose $n = 1$ since $a > 0 \geq b$.

(h) False: take $a = -1$, so $-a = 1$ is positive.

(i) False: at $a = 0$, neither a nor $-a$ is positive. One is positive when $a \neq 0$.

(j) True: $1, 2 \cdot 1, 3 \cdot 1, \ldots$ are distinct because a positive sum of 1's cannot be zero.

Exercise 25.16: Making pi larger than every rational

Since π is transcendental, evaluation $\phi_\pi : \mathbb{Q}[x] \to \mathbb{Q}[\pi]$ is an isomorphism. Give $\mathbb{Q}[x]$ the high-degree ordering and transfer it through this map. Equivalently a nonzero $a_0 + a_1\pi + \cdots + a_n\pi^n$ is positive exactly when its highest nonzero rational coefficient is positive. Then for every $q \in \mathbb{Q}$, $\pi - q$ has highest coefficient 1, so $\boxed{q < \pi}$ in this ordering. Transcendence ensures a unique polynomial representation, so the sign rule is well defined.

Exercise 25.17: Conjugation of a quadratic ring

Representations $m + n\sqrt{2}$ are unique by irrationality of $\sqrt{2}$. For $u = m + n\sqrt{2}$, $v = r + s\sqrt{2}$, direct calculation gives

$$\phi(u + v) = (m + r) - (n + s)\sqrt{2} = \phi(u) + \phi(v).$$

Since $uv = (mr + 2ns) + (ms + nr)\sqrt{2}$,

$$\phi(uv) = (mr + 2ns) - (ms + nr)\sqrt{2} = (m - n\sqrt{2})(r - s\sqrt{2}).$$

Thus multiplication is preserved as well. In fact ϕ^2 is the identity, so the homomorphism is an automorphism.

Exercise 25.18: Cone membership means greater than zero

By the definition in Theorem 25.5, $0 < a$ means exactly $a - 0 \in P$. This equals $a \in P$, which is the hypothesis. Thus the conclusion follows directly from the definition of $<$.

Exercise 25.19: Equal positive multiples

First, an ordered ring has no nonzero zero divisors: for nonzero u, v, choose signs so that $\pm u, \pm v \in P$; closure makes one of $uv, -uv$ positive, so $uv \neq 0$. Hence if $c = 0$, then $bd = 0$ with $b \neq 0$ forces $d = 0$, and similarly with $d = 0$.

Suppose now that neither is zero. If $c \in P$, then $ac \in P$. If $-d \in P$, closure would also give $b(-d) = -bd = -ac \in P$, contradicting trichotomy. Thus $d \in P$, and $cd \in P$. If instead $-c \in P$, the same argument applied to $a(-c) = b(-d)$ gives $-d \in P$, whence $cd = (-c)(-d) \in P$. This proves exactly the stated alternatives using only cone closure and trichotomy.

Exercise 25.20: Reversing signs reverses an inequality

The hypothesis $a < b$ means $b - a \in P$. But

$$(-a) - (-b) = b - a \in P.$$

By the definition of $<$, this is precisely $-b < -a$.

Exercise 25.21: A negative times a positive

From $a < 0$ we obtain $-a \in P$, and from $0 < b$ we obtain $b \in P$. Closure under multiplication gives $(-a)b = -ab \in P$. Since $0 - ab = -ab$ is positive, the defining condition for $<$ gives $ab < 0$. No separate sign rule has been assumed.

Exercise 25.22: A positive quotient

Since $b > 0$, it is nonzero and b^{-1} exists. The square of any nonzero element is in P: trichotomy makes either u or $-u$ positive, and squaring that element gives $u^2 \in P$. Thus $(b^{-1})^2 \in P$. Multiplying by $b \in P$ gives $b^{-1} = (b^{-1})^2 b \in P$. Finally closure gives $a/b = ab^{-1} \in P$, proving positivity of the quotient.

Exercise 25.23: Inverting a positive element below one

The assumptions give $a \in P$ and $1 - a \in P$. By Exercise 25.22, $(1 - a)/a \in P$. But this quotient is $a^{-1} - 1$. Therefore $1 < a^{-1}$ by the defining relation $<$.

Exercise 25.24: Inverting a negative element above minus one

Let $b = -a$. The inequalities $-1 < a < 0$ imply $0 < b < 1$, using the difference definition of $<$ or Exercise 25.20. Exercise 25.23 gives $1 < b^{-1}$. Reversing signs yields $-b^{-1} < -1$. Since $a^{-1} = (-b)^{-1} = -b^{-1}$, the conclusion is $a^{-1} < -1$.

Exercise 25.25: Transporting an ordering across an isomorphism

Let $P' = \phi[P]$. It is nonempty because P is nonempty. If $u = \phi(a)$ and $v = \phi(b)$ with $a, b \in P$, then $u + v = \phi(a + b) \in P'$ and $uv = \phi(ab) \in P'$ by closure of P. For any $u \in R'$, take its unique preimage a. Exactly one of $a \in P$, $a = 0$, $-a \in P$ holds, and injectivity and surjectivity translate this into exactly one of $u \in P'$, $u = 0$, $-u \in P'$. Hence P' is an ordering cone. Finally

$$\phi(a) <' \phi(b) \iff \phi(b) - \phi(a) = \phi(b - a) \in \phi[P] \iff b - a \in P \iff a < b.$$

The middle reverse implication uses injectivity of ϕ. This proves Theorem 25.10 in full.

Exercise 25.26: The induced ordering on a subring

For a nonzero subring S, put $P_S = P \cap S$. It is nonempty: choose $s \in S \setminus \{0\}$, and trichotomy gives either s or $-s$ in P_S. Sums and products of elements of P_S lie in P by cone closure and in S by the

subring axioms. For any $s \in S$, exactly one of $s \in P$, $s = 0$, $-s \in P$ holds; since both $s, -s$ lie in S, this is exactly the cone trichotomy for P_S. Thus the induced order is the restriction of the original order.

The nonzero qualification matters: for $S = \{0\}$ the intersection is empty, contrary to the definition's nonempty-cone requirement. If one also imposes the section's standing requirement that every ordered ring have nonzero unity, restrict the assertion to subrings with nonzero unity. For nonunital nonzero subrings the same proof still supplies the asserted order and cone axioms.

Exercise 25.27: Recovering the cone from the order relation

Set $P = \{x : 0 < x\}$. Since the ring is nonzero, compare 0 with some $a \neq 0$. Trichotomy gives either $0 < a$ or $a < 0$; in the latter case additive isotonicity gives $0 < -a$. Thus P is nonempty. If $a, b \in P$, adding a to $0 < b$ gives $a < a + b$, and transitivity with $0 < a$ gives $0 < a + b$. Multiplicative isotonicity applied to $0 < b$ and $0 < a$ gives $0 = a0 < ab$. So P is closed under both operations.

Trichotomy of 0 and x, and additive isotonicity translating $x < 0$ to $0 < -x$, give exactly one of $x \in P$, $x = 0$, $-x \in P$. Thus this is a positive cone. Finally additive isotonicity, in both directions, gives

$$a < b \iff 0 < b - a \iff b - a \in P.$$

Therefore the relation induced by P is the original relation, proving the converse in Theorem 25.5.

Exercise 25.28: Injectivity of an odd power in an ordered domain

An odd power has the same sign as its nonzero base: a positive base gives a positive product; for a negative base, its negative is positive and $(-c)^{2n+1} = -c^{2n+1}$. Thus equal odd powers cannot come from bases of opposite signs, or from one zero and one nonzero base.

For positive $a < b$, factor

$$b^{2n+1} - a^{2n+1} = (b - a)\big(b^{2n} + b^{2n-1}a + \cdots + a^{2n}\big).$$

Every summand in the parenthesis is positive, so the difference is positive and equality of the powers is impossible. The case $0 < a = b$ already has the conclusion. If $a, b < 0$, their negatives are positive, and equality of odd powers gives equality of the odd powers of $-a, -b$, reducing to the positive case. Trichotomy therefore leaves only $\boxed{a = b}$.

Exercise 25.29: Eight different orderings in two variables

For each stage choose low (L) or high (H) degree to determine the sign. Applying the univariate construction twice gives eight orderings. To distinguish them, record whether four fixed polynomials

are positive (+) or negative (−):

first ring, then outer variable	inner	outer	$x-1$	$y-1$	$x-y$	$xy-1$
$(R[x])[y]$	L	L	−	−	+	−
$(R[x])[y]$	L	H	−	+	−	+
$(R[x])[y]$	H	L	+	−	+	−
$(R[x])[y]$	H	H	+	+	−	+
$(R[y])[x]$	L	L	−	−	−	−
$(R[y])[x]$	L	H	+	−	+	+
$(R[y])[x]$	H	L	−	+	−	−
$(R[y])[x]$	H	H	+	+	+	+

.

For example, with $(R[x])[y]$ and outer L, the deciding y^0 coefficient of $x-y$ is $x>0$, while that of $xy-1$ is $-1<0$. With outer H, their deciding coefficients are -1 and x, reversing those two signs. The other rows follow by interchanging x, y and choosing the specified inner signs for $x-1$ or $y-1$. No two sign rows agree. Thus no two of the eight orderings can be the same.